



PERFORMANCE ANALYSIS OF CONVOLUTIONAL NEURAL NETWORK ARCHITECTURES FOR IMAGE CLASSIFICATION

Arthur Dohnert, Eduardo Menezes,
Fernando Tedesco

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- **Problem:** Deploying CNNs in resource-constrained environments (Edge, Embedded, Shared GPU Clusters).
- **Trade-off:** High predictive performance vs. low latency and memory usage.
- **Goal:** Systematically benchmark lightweight architectures under a unified, reproducible pipeline.

Light Models (Green)

- MobileNet V1
- ShuffleNet V2
- SqueezeNet
- EfficientNet-B0

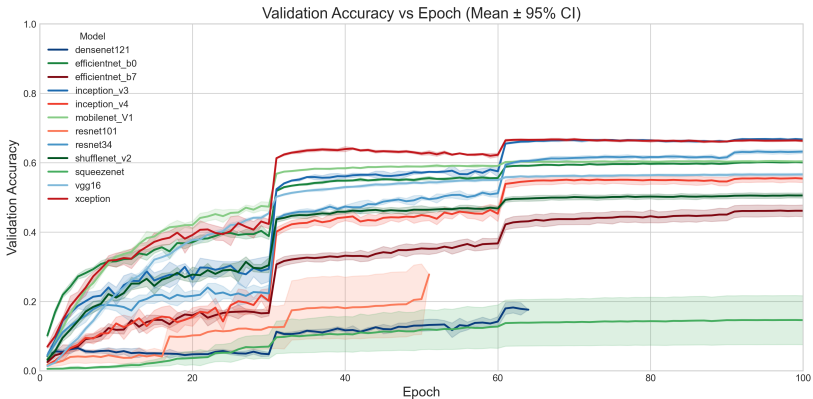
Medium Models (Blue)

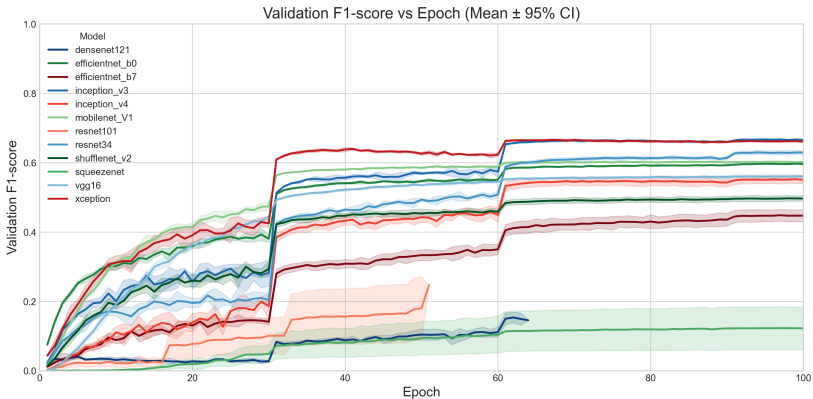
- VGG16
- ResNet34
- Inception V3
- DenseNet121

Heavy Models (Red)

- ResNet101
- Inception V4
- Xception
- EfficientNet-B7

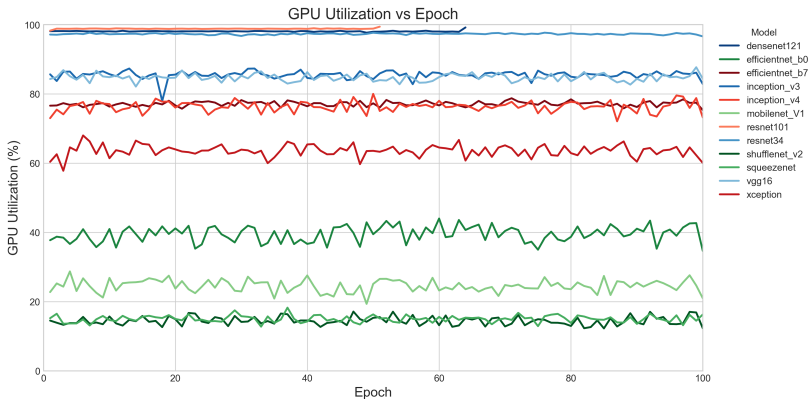
- **Hardware:** UFRGS PCAD Cluster (Tupi partition).
 - Single NVIDIA GPU (24 GiB VRAM) per job.
 - Data staged to node-local SCRATCH to minimize I/O latency.
- **Software:**
 - Python 3.11, PyTorch 2.0.
 - PYTORCH_CUDA_ALLOC_CONF tuned to reduce fragmentation.
- **Dataset:** Tiny ImageNet (200 classes, 64×64 px).
- **Protocol:** 100 Epochs, 5 Random Seeds per model.

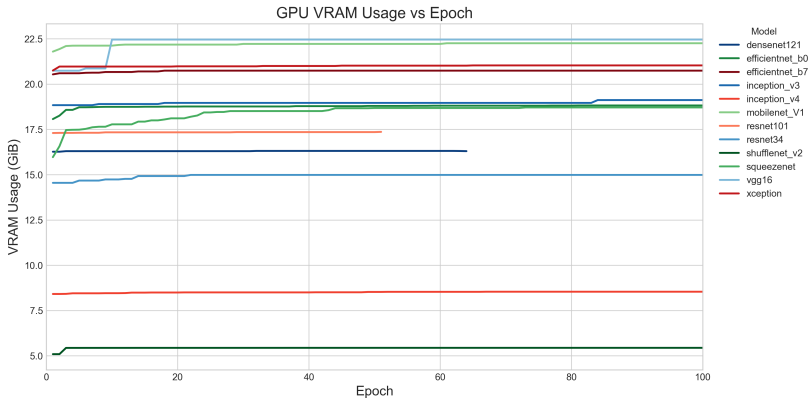


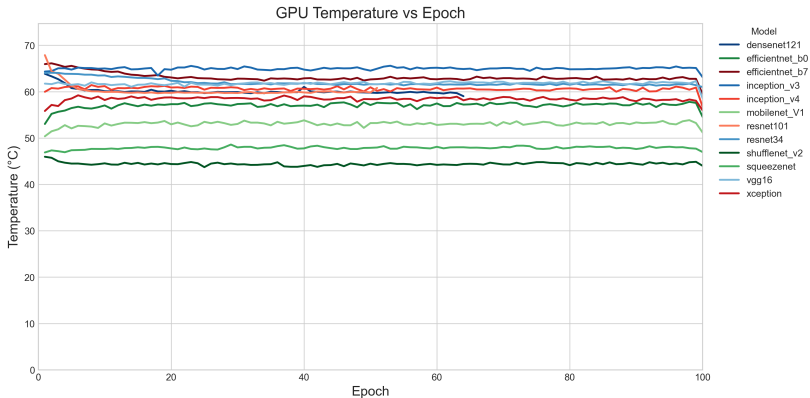


Model	Top-1 Accuracy	F1-Score
DenseNet121	0.184 ± 0.015	0.154 ± 0.015
EfficientNet-B0	0.603 ± 0.002	0.598 ± 0.002
EfficientNet-B7	0.463 ± 0.017	0.449 ± 0.018
Inception V3	0.670 ± 0.002	0.669 ± 0.002
Inception V4	0.557 ± 0.011	0.553 ± 0.011
MobileNet V1	0.606 ± 0.002	0.604 ± 0.002
ResNet101	0.207 ± 0.102	0.181 ± 0.092
ResNet34	0.633 ± 0.005	0.630 ± 0.005
ShuffleNet V2	0.506 ± 0.008	0.498 ± 0.008
SqueezeNet	0.147 ± 0.071	0.123 ± 0.062
VGG16	0.567 ± 0.004	0.561 ± 0.004
Xception	0.669 ± 0.003	0.667 ± 0.003

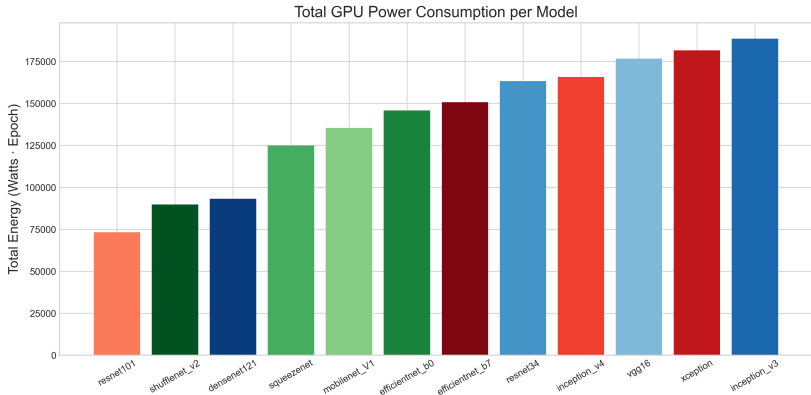
Tabela: Peak metrics (Mean $\pm$ 95% CI) across 5 seeds.

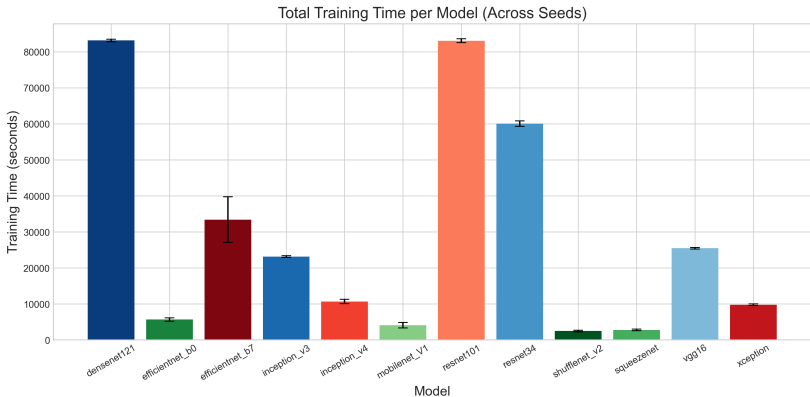


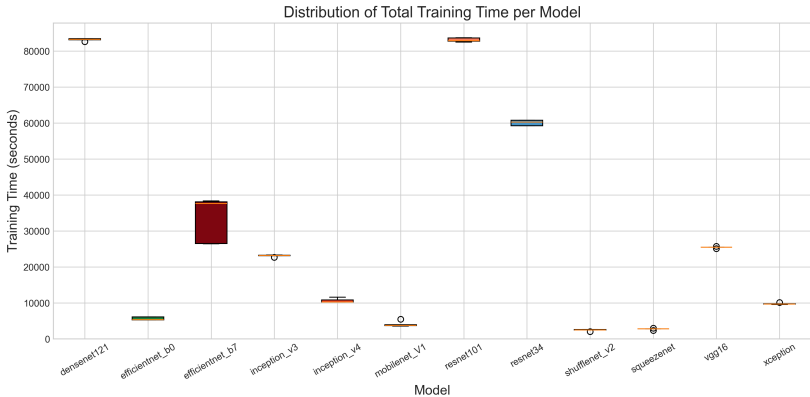


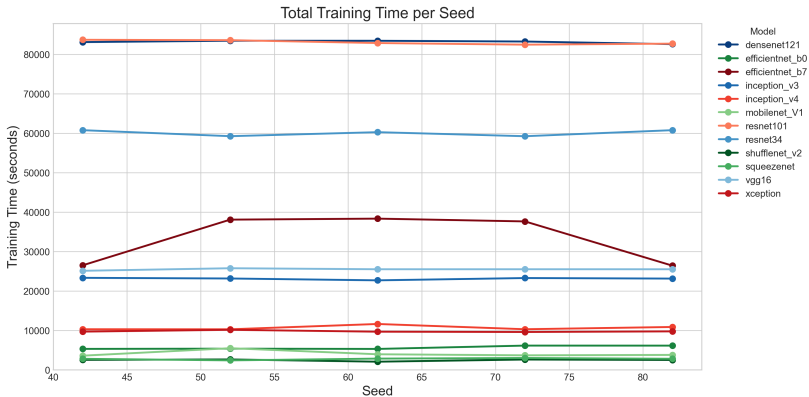


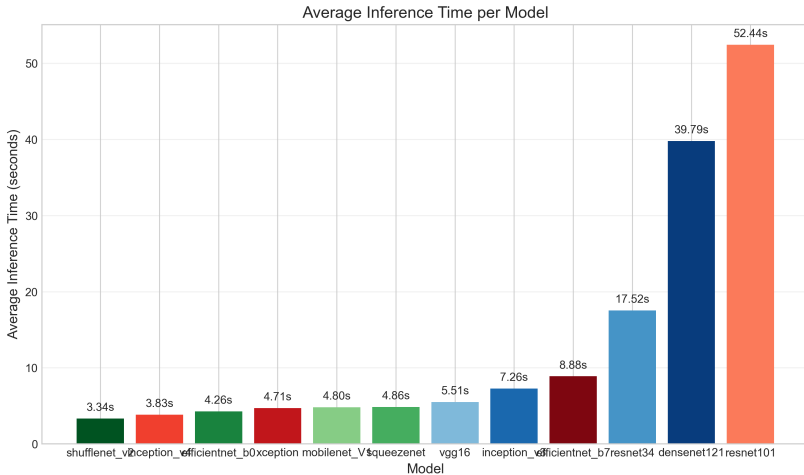
TOTAL ENERGY CONSUMPTION PER MODEL

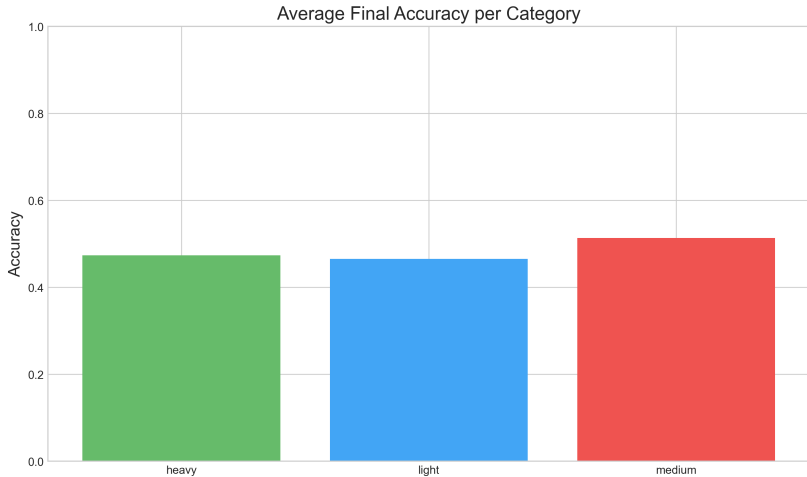




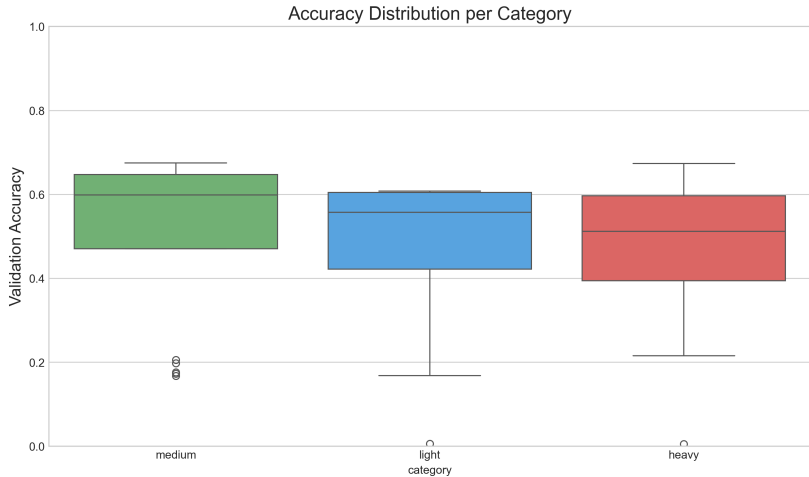


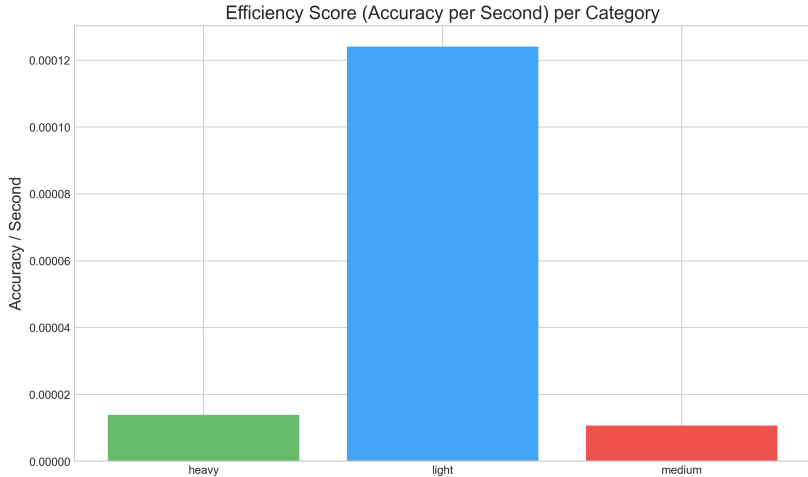


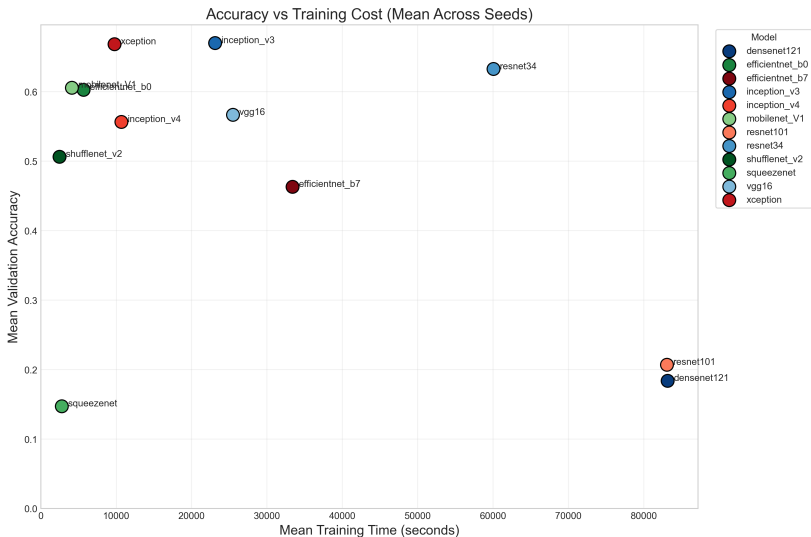


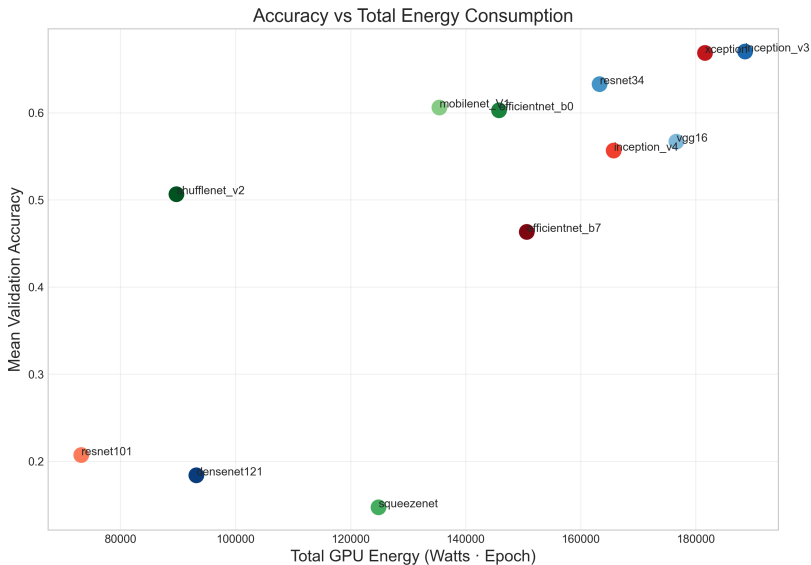












1. **Performance does not strictly follow model size.**
Surprisingly, several heavy models underperformed compared to lightweight architectures. Inception V3, ResNet34, MobileNet V1, and Xception achieved the best accuracies, despite having very different computational footprints.
2. **Training time was also inconsistent with model complexity.** Some lightweight models (e.g., SqueezeNet) trained slower than expected, while certain heavier models (e.g., Inception V3, Xception) trained efficiently.

- **Fixed Hyperparameters.** All models were trained using identical hyperparameters. This design isolates architectural effects but likely underestimates the true performance of models that require tailored tuning (e.g., SqueezeNet, EfficientNet).
- **Hardware Specificity.** Results reflect performance on NVIDIA GPUs. Latency and efficiency trends may differ on CPU-only or low-power edge devices where memory patterns dominate.
- **Intrinsic Architectural Constraints.** Some architectures carry structural limitations that hinder convergence under standardized training settings. Examples include SqueezeNet's aggressive compression, DenseNet's feature-map accumulation, and the depth of large ResNet variants. These models may require architecture-specific schedules to reach full potential.

**Arthur Dohnert, Eduardo Menezes,
Fernando Tedesco**
Instituto de Informática — UFRGS

